



# Koneru Lakshmaiah Education Foundation

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## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

|                             |                                                                                      |                       |
|-----------------------------|--------------------------------------------------------------------------------------|-----------------------|
| <b>TITLE OF THE PROJECT</b> | AssetFlow: IT Asset & Inventory Management System give modern tools and technologies |                       |
| <b>S. No.</b>               | <b>University ID</b>                                                                 | <b>Name</b>           |
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## Abstract

**AssetFlow: IT Asset & Inventory Management System** is a modern, centralized web-based platform developed to simplify and automate the management of an organization's IT assets and inventory. In many organizations, tracking devices such as laptops, desktops, monitors, printers, networking equipment, and accessories using spreadsheets or manual records can result in data inconsistency, asset loss, duplicate entries, and difficulty in monitoring asset availability. AssetFlow addresses these challenges by providing a structured system for managing the complete lifecycle of IT assets.

The system allows administrators and authorized users to add, update, categorize, assign, transfer, and track assets efficiently. Each asset can be associated with important information such as asset ID, category, purchase details, assigned employee, location, condition, availability, and current status. The system also maintains asset allocation and return records, enabling organizations to easily identify who is using an asset, where it is located, and whether it is available, assigned, under maintenance, or retired. A centralized dashboard can provide useful insights into total assets, available assets, assigned assets, maintenance requirements, and inventory status.

The frontend of AssetFlow is developed using **React.js**, providing a responsive and user-friendly interface for interacting with the system. **Spring Boot 3** is used to develop the backend and RESTful APIs, providing a scalable and modular architecture for handling business logic, authentication, asset operations, and communication between the frontend and database. **PostgreSQL** is used as the relational database for securely storing asset, user, assignment, and inventory information.

For deployment and environment consistency, **Docker** is used to containerize the application and its supporting services. **GitHub** is used for source-code management, version control, collaboration, and maintaining the development history. **Jira** is integrated into the development workflow to support Agile practices such as sprint planning, task tracking, issue management, and progress monitoring.

Signature of the Guide

HOD